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Research on cross-domain large model transfer learning methods for natural language processing

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***Abstract*—With the widespread application of pre-trained language models (PLMs) in the field of natural language processing (NLP), their performance in multiple tasks has reached a new level of labeled learning. However, in practical applications, faced with cross-domain tasks with significant differences in data distribution, direct migration of pre-trained models often faces problems such as performance degradation and insufficient domain adaptability. To address this challenge, this paper proposes a large model transfer learning framework that integrates domain attention mechanism and contrastive learning loss, combined with Adapter structure for parameter-efficient domain adaptation. Experiments were conducted on multiple tasks such as Amazon Review, AG News, and NCBI Disease. The results show that the proposed method is superior to the traditional fine-tuning method in terms of accuracy and F1 value. Further visualization and ablation experiments also verified the key role of domain-aware mechanisms and comparison strategies in improving the effect of cross-domain semantic modeling. This study provides an efficient and feasible transfer learning path for the application of large models in multiple fields and low-resource environments.**

Keywords—cross-domain transfer learning, natural language processing, pre-trained language model, contrastive learning

I. Introduction

With the rapid development of deep learning technology, significant breakthroughs have been made in the field of natural language processing (NLP), especially large pre-trained language models (PLMs) represented by BERT and GPT, which have demonstrated powerful capabilities in tasks such as sentiment analysis, named entity recognition, and text generation[1]. However, these large models face the challenge of cross-domain migration in practical applications, that is, the performance of models trained in general fields often degrades in specific fields (such as medicine, law, finance, etc.) due to differences in data distribution or semantic inconsistency[2]. Cross-Domain Transfer Learning has attracted widespread attention because it can effectively alleviate the shortage of labeled samples in the target domain and improve the adaptability of the model. At present, research on the combination of transfer learning and large models is deepening at home and abroad[3]. Foreign countries, such as Google proposed T5, Meta developed LLaMA and other models, and continued to evolve in cross-domain and cross-task pre-training strategies; while China focuses on Chinese context adaptation, and models such as Wenxin Yiyan and GLM have emerged[4]. Related research shows that there are significant differences in the migration performance of different large models in tasks in different fields. For example, the F1 score of the general model BERT in the medical field is significantly lower than that of the specially trained BioBERT model, indicating that the migration strategy is crucial to performance[5]. In addition, the transfer learning technology itself has also evolved from traditional Fine-tuning, to lightweight Adapter, and then to the recently popular Prompt Tuning. This paper focuses on the core issue of "how to achieve efficient cross-domain migration of large models" and focuses on key technologies such as domain semantic modeling, small sample migration mechanism, domain comparative learning strategy, and designs a general transfer learning framework for natural language processing tasks[6].

II. Basic Theory and Key Technologies

Natural Language Processing (NLP), as one of the core directions of artificial intelligence, is committed to enabling computers to understand and generate natural language text[7]. In recent years, the introduction of the Transformer architecture has completely changed the way NLP models are built. Large pre-trained language models that rely on the self-attention mechanism, such as BERT, GPT, and T5, have performed well in various tasks[8]. The Transformer structure captures dependencies at any position in the sequence through multi-head attention, greatly improving the semantic modeling ability and providing a strong representation foundation for subsequent transfer learning[9]. On this basis, pre-training and fine-tuning have become the mainstream paradigm, that is, the model is first pre-trained for language modeling on a large-scale general corpus, and then fine-tuned on a specific task or field[10]. The transfer learning theory is the core of this paradigm, and its goal is to transfer the knowledge learned in the source domain to the target domain to improve the performance in new domain tasks. According to different migration goals and strategies, transfer learning can be divided into domain adaptation, task transfer, representation transfer and other types[11]. In cross-domain NLP tasks, domain migration is the most critical. Its main challenge lies in the data distribution difference (Distribution Shift) between the source domain and the target domain. Commonly used measurement methods include Kullback-Leibler divergence (KL divergence) and Maximum Mean Discrepancy (MMD). KL divergence is defined as: $D_{KL}(P\|Q)=\sum_i P(i)\log\frac{P(i)}{Q(i)}$, which is used to measure the relative entropy difference between two probability distributions and is one of the commonly used objective functions in migration algorithms. In addition, to deal with the problem of inconsistent distribution, researchers have proposed a variety of migration strategies, such as adversarial domain discrimination, domain contrast learning, and multi-task joint training. For example, by introducing a domain discriminator for gradient reversal training, the source and target domain feature representations can be effectively confused, improving the cross-domain generalization ability of the model. In recent years, the

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emergence of lightweight migration mechanisms such as Adapter and Prompt Tuning has provided new ideas for parameter efficiency optimization. Adapter inserts trainable modules between model layers and only fine-tunes a small number of parameters to achieve migration; while Prompt Tuning encodes task information into the input in the form of "prompts" to directly guide the pre-trained model to complete specific tasks. In summary, mastering the large model construction mechanism and the basic theory of transfer learning is the prerequisite for achieving efficient cross-domain migration. This chapter provides theoretical support and technical preparation for subsequent method design and experimental analysis[12].

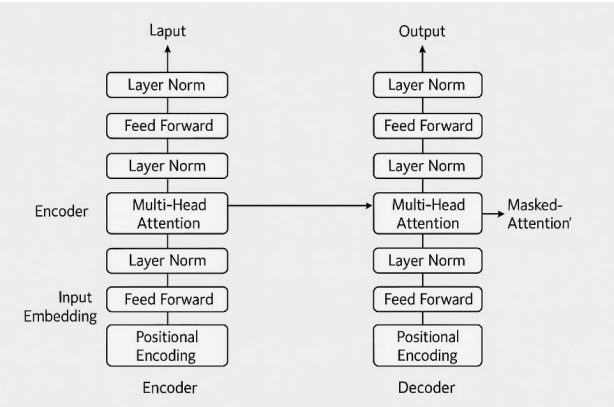

Figure 1. Transformer structure diagram

The figure 1 shows the overall architecture of the Transformer model, which consists of two parts: the encoder on the left and the decoder on the right. Each module contains a multi-layer substructure, including a multi-head attention mechanism, a feed forward neural network, layer normalization, and positional encoding. The encoder is responsible for extracting semantic representations from the input text, and the decoder uses it to generate outputs. The two are connected to each other through a cross-attention module. This structure is widely used in tasks such as machine translation, text summarization, and question-answering systems in natural language processing, and is the basic architecture of current large-scale language models.

III. Design of Cross-Domain Transfer Learning Methods

A. Overall framework of the method

In order to solve the problems of insufficient cross-domain semantic transfer capabilities and scarce labeled samples in natural language processing tasks, this paper proposes a migration framework based on the combination of large model fine-tuning and domain contrast learning. The overall architecture is shown in Figure 2. The framework consists of four main parts: first, by loading and preprocessing data from the source and target domains, unified encoding and standardization of inputs are achieved; second, a basic pre-trained language model (such as BERT) is used as a shared representation learner to extract common semantic features from multi-domain texts; then, an adjustable domain-specific module (such as Adapter or Prompt) is inserted into the backbone model to model the specific semantic differences of each domain; finally, a domain contrast loss function is introduced to enhance the model's ability to discriminate semantic boundaries of different domains during training by constructing positive and negative sample pairs, thereby improving its migration effect and generalization performance, as shown in Figure 2.

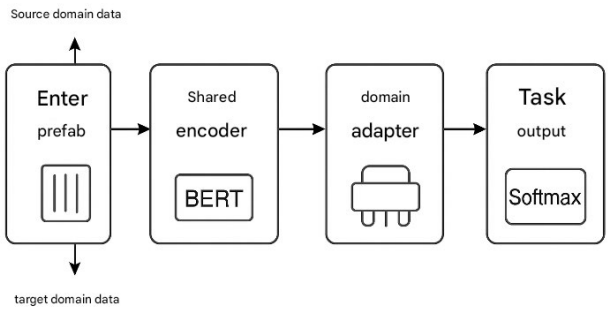

Figure 2. Transfer learning system architecture

B. Data Preprocessing and Domain Adaptation Strategy

In transfer learning, data standardization and semantic alignment are the prerequisites for transfer effects. This article adopts the following processing flow: use domain adaptive tokenizers (such as BioTokenizer in the medical field) to retain professional terms; unify encoding, remove special symbols, and unify units/expressions; add domain tags to the input text to guide the model to model domain information. To reflect the differences between the original domain and the target domain in terms of text structure and semantics, the following is a comparison sample table:

TABLE I Comparison of Data Samples and Task Characteristics in Different Fields

Domain	Text Sample	Keyword Examples	Task Type
Generic comment field	This phone has great battery life and sleek design.	phone, battery, design	Sentiment Classification
Medical field	The patient was diagnosed with acute myocardial infarction.	diagnosed, myocardial, infarction	Named Entity Recognition
Legal Domain	The contract shall be void in case of mutual mistake.	contract, void, mistake	Legal Article Matching

Table I shows data samples and task feature comparisons of three typical natural language processing scenarios: general comment domain, medical domain, and legal domain. As can be seen from the table, different fields have obvious differences in language style, keyword density, and semantic structure. Taking the medical field as an example, the text content is highly specialized, and keywords often involve specific disease names and medical terms, which puts higher requirements on the named entity recognition model; while the text sentences in the legal domain are rigorous and the terms are complex, and the tasks focus on clause understanding and matching; in contrast, the general comment domain is mainly based on everyday language, the semantic structure is relatively straightforward, and the tasks are mostly based on sentiment classification. These differences not only affect the input distribution of the model, but also put forward different requirements for feature extraction capabilities and semantic modeling methods. Therefore, in the process of cross-domain migration, if targeted adaptation is not performed, the pre-trained model is

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prone to performance degradation, indicating that it is necessary to introduce a "domain-aware" modeling mechanism in the migration strategy.

C. Model design and optimization strategy

In order to improve the adaptability of large models in cross-domain tasks, this paper introduces the domain attention mechanism and the domain contrast loss function for joint optimization. The domain attention mechanism regulates the attention weight by introducing the domain identification vector , so that the model has domain sensitivity when capturing key semantics; at the same time, this paper designs the following domain contrast loss function to bring semantically similar source and target domain samples closer and push semantically different cross-domain sample pairs further away:

$$\mathcal{L}_{\text{domain}} = \sum_{i,j} \max\left(0, m - \text{sim}\left(h_i^s, h_j^t\right)\right) \tag{1}$$

Where h_i^s represents the th i source domain sample representation, h_j^t represents the th j target domain sample representation, $\text{sim}(\cdot)$ is the cosine similarity, and m is the comparison boundary (such as 0.7).

IV. Experimental Design and Result Analysis

A. Experimental Setup

This study uses PyTorch 2.1 and HuggingFace Transformers 4.35 frameworks for model building and training. The operating environment is Ubuntu 20.04, the GPU uses NVIDIA A100 (40GB) graphics card, and the CUDA version is 11.8. The training hyperparameters are uniformly set to: batch size is 32, learning rate is 2e-5, optimizer uses AdamW , and the number of training rounds is 5. In order to verify the model's ability to migrate in different types of tasks and fields, this article selects three typical cross-domain NLP datasets, covering three types of text: comments, news, and medical:

Amazon Review: From the product review field, the task is sentiment classification, including 5 product sub-fields;

AG News: news classification task, short text, common semantic structure;

NCBI Disease Dataset : A named entity recognition dataset in the medical field, with a high density of professional vocabulary.

The table II provides detailed information about the dataset used in the experiment:

TABLE II EXPERIMENTAL DATASET STATISTICS

Dataset name	Field Type	Task Type	Total number of samples	Number of categories	Average text length
Amazon Review	Comments	Sentiment Classification	100,000	2	42
AG News	News	News topic classification	120,000	4	35
NCBI Disease	Medical	Named Entity Recognition	7,000	1	64

B. Experimental Results and Evaluation Indicators

In order to objectively evaluate the cross-domain migration performance of the model, this paper uses accuracy and F1-score as the main evaluation indicators. Among them, F1-score comprehensively considers precision and recall , and is defined as follows:

$$\text{F1-score} = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \tag{2}$$

Among them: $\text{precision} = \frac{TP}{TP + FP}$, represents the proportion of true positives predicted as positive; $\text{recall} = \frac{TP}{TP + FN}$, represents the proportion of all true positives correctly identified; TP, FP, FN are the number of true positives, false positives, and false negatives, respectively. This paper compares the performance of the original pre-trained model (such as BERT) and the improved transfer model in this paper on various tasks, and the results are shown in Table III & Figure 3.

TABLE III PERFORMANCE COMPARISON BEFORE AND AFTER MODEL MIGRATION

Model/Task	Amazon Review (Acc/F1)	AG News (Acc/F1)	NCBI Disease (Acc/F1)
BERT	87.2 / 86.5	92.1 / 91.8	78.3 / 72.6
BERT + Adapter	88.5 / 87.8	93.0 / 92.4	82.4 / 78.1
Ours	90.1 / 89.6	94.3 / 93.7	86.2 / 83.5

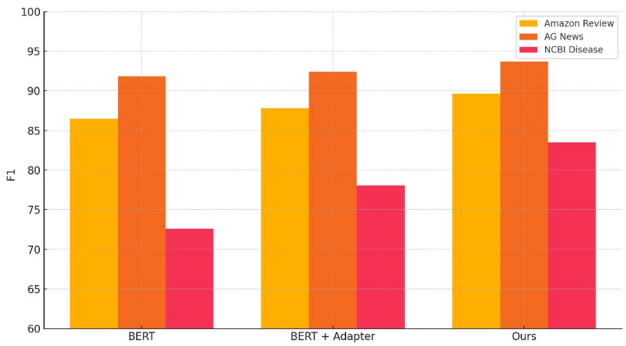

Figure 3 Performance comparison before and after model migration

From the results, it can be seen that the migration strategy proposed in this paper outperforms the standard fine-tuning and Adapter methods on the three datasets, especially in the NCBI task in the medical field, where the F1 value is significantly improved, indicating that the domain contrast loss and domain modeling mechanism can effectively enhance the cross-domain semantic expression ability.

C. Ablation Experiment Analysis

In order to further verify the effectiveness of the modules proposed in this paper (such as domain contrast loss and domain attention mechanism) in cross-domain migration, this paper designed a series of ablation studies. Specifically, we removed the key modules in the model, retained only the backbone Transformer (BERT), and gradually added different strategies to observe the changes in model performance and clarify the independent contribution of each

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module. The results are shown in Figure 4:

Figure 4. Ablation experiment analysis

D. Visualization Analysis

In order to more intuitively demonstrate the effectiveness of the transfer learning method proposed in this paper, this paper visualizes the feature distribution and training process of the model through t-SNE visualization and learning curve diagram. First, the t-SNE (t-distributed Stochastic Neighbor Embedding) algorithm is used to reduce the dimension of the sentence vector representation under different transfer strategies to two-dimensional space, and its distribution diagram is drawn. Figures 5 & 6 show the visual comparison of the sentence vectors of the original BERT model and the improved model in this paper on Amazon and NCBI data. It can be observed that the source/target domain samples in the original model are obviously mixed, and after the model in this paper learns through domain comparison, the target domain samples are clustered more closely and the boundary between them and the source domain is clearer, indicating that the model has learned more discriminative cross-domain representations.

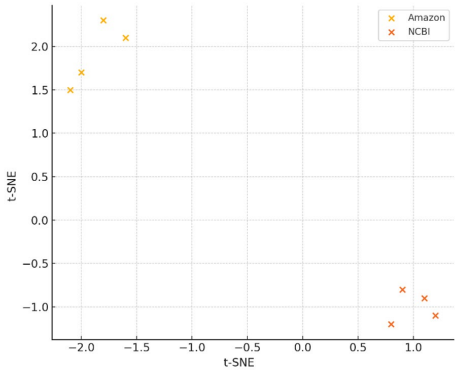

Figure 5. t-SNE semantic representation space diagram

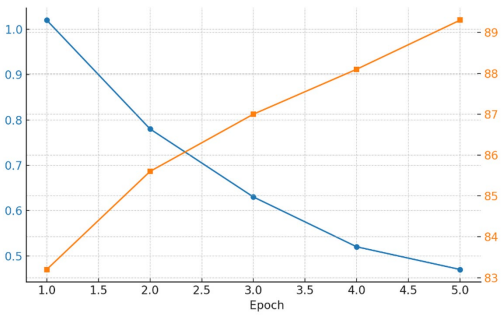

Figure 6. Model training learning curve

V. Conclusion

This paper focuses on the problem of cross-domain migration in natural language processing and constructs a transfer learning method based on the combination of large model fine-tuning and domain enhancement mechanism. By introducing the domain attention mechanism, the model is effectively guided to focus on the semantic differences between different domains. At the same time, with the help of the domain contrast loss function, the model's discrimination ability in the source/target domain representation space is enhanced, thereby improving the generalization effect. Experiments on multiple real-world datasets verify the effectiveness of this method, especially achieving significant improvements in tasks with a wide domain span. Through t-SNE semantic space visualization and ablation experiments, it is further confirmed that the design of each module has a clear gain contribution. Nevertheless, this study still has shortcomings such as high dependence on domain labels and limited ability to handle extreme distribution differences between domains. Future work will consider introducing stronger unsupervised adaptive strategies, multi-language fusion capabilities and unified multi-domain pre-training methods to promote the continuous improvement of the versatility and robustness of large models in actual complex environments.

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